

IRFAN UD DIN

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CAREER OBJECTIVE

Motivated BS Computer Science student with a CGPA of 3.42/4.00, seeking an internship opportunity in Artificial Intelligence, Machine Learning or AI Backend. Eager to apply my skills in Python, machine learning, backend development, databases, and problem solving to real-world projects while gaining practical industry experience and improving my technical expertise.

EDUCATION

NUML, Islamabad
Bachelor of Science in Computer Science
CGPA: 3.42 / 4.00
Jan 2024 - Dec 2027

The Langlands School and College, Chitral
FSC Pre-Engineering
2020 – 2022

TECHNICAL SKILLS

Programming Languages: Python, C++, C#, JavaScript, HTML, CSS

Machine Learning and Data: Machine Learning, Linear Regression, Data Preprocessing, Feature Engineering, Regression Modeling, NumPy, pandas, Matplotlib, scikit-learn

Backend Development: Node.js, Express.js, REST APIs, JWT Authentication, Role-Based Access Control, API Validation

Frontend Development: React, HTML, CSS, Streamlit

Databases: PostgreSQL, MySQL

Frameworks and Tools: Streamlit, TypeORM, ASP.NET MVC, .NET Framework, WinForms, joblib, GitHub

Core Computer Science: Object-Oriented Programming, Data Structures and Algorithms, Problem Solving, Debugging, Authentication, Mathematics, Statistics

Soft Skills: Communication

PROJECTS

House Price Prediction System

Technologies: Python, Machine Learning, scikit-learn, pandas, NumPy, Matplotlib, Streamlit, joblib

Live Demo: <https://house-price-prediction-pakistan-iub.streamlit.app/>

- Built a machine learning-based web application to predict house prices in Pakistan using a real estate dataset of 168,446 records.
- Cleaned and prepared housing data by selecting important features such as area, bedrooms, bathrooms, city, location, and price.
- Converted property area values from Marla and Kanal into square feet to create a consistent numerical feature for model training.
- Implemented a scikit-learn machine learning pipeline using ColumnTransformer, SimpleImputer, OneHotEncoder, and Linear Regression.
- Applied median imputation for numerical features and one-hot encoding for categorical features such as city and location.
- Saved the trained machine learning model using joblib for reuse inside the web application.
- Developed a Streamlit user interface that allows users to enter area, bedrooms, bathrooms, city, and location to estimate house price.

LCM Sports E-Commerce Backend

Technologies: Node.js, Express.js, PostgreSQL, TypeORM, JWT, bcryptjs, Multer, Jest, Supertest

- Developed a RESTful e-commerce backend API for a sports products platform using Node.js, Express.js, PostgreSQL, and TypeORM.
- Designed database entities for users, products, categories, carts, cart items, orders, order items, payments, reviews, and wishlists.
- Implemented JWT-based authentication with protected routes and role-based access control for admin, seller, and customer users.
- Built API modules for user authentication, product management, categories, cart, orders, payments, reviews, wishlist, and dashboards.
- Added backend validation using express-validator for registration, login, products, and order-related requests.
- Implemented product image upload functionality using Multer with file type filtering and file size limits.
- Added automated API tests using Jest and Supertest for authentication, product, and order features.

LinkedIn Clone

Technologies: HTML, CSS

- Designed a responsive LinkedIn clone focusing on profile layout, navigation bar, and post feed interface.
- Applied modern UI design principles and responsive web layout techniques to improve page structure and usability.

LANGUAGES

Urdu
English